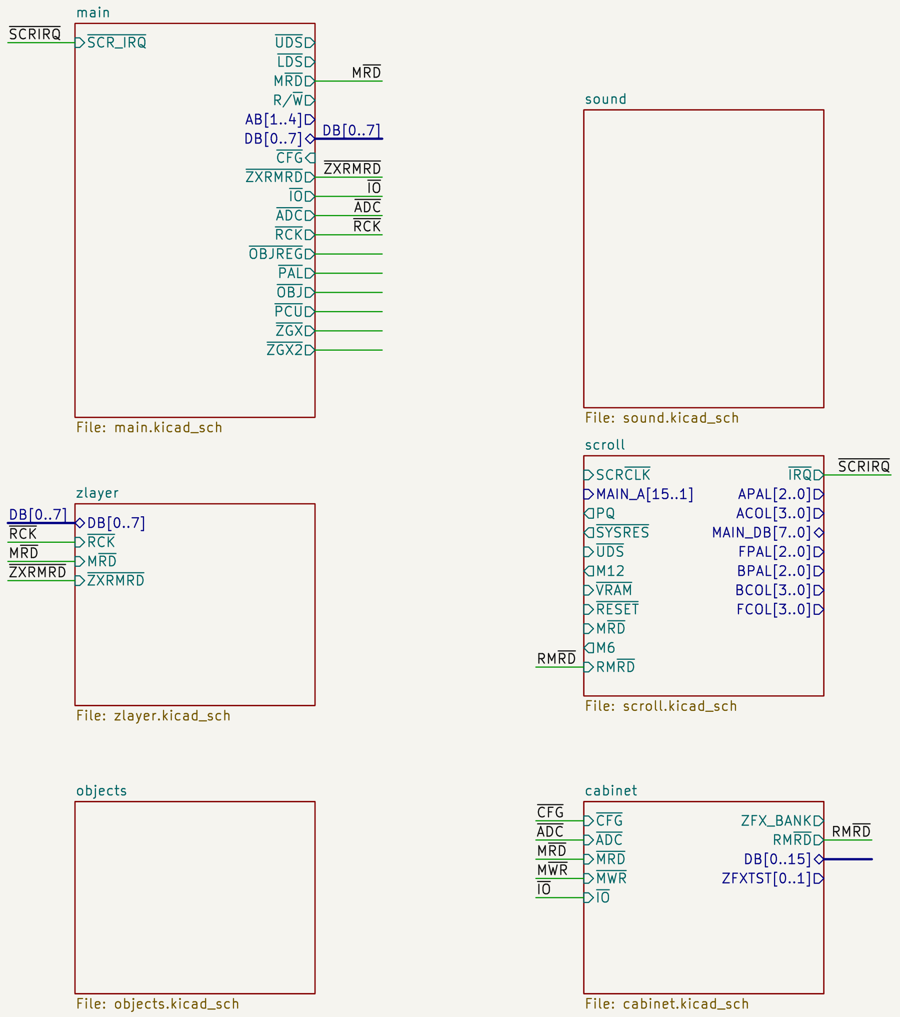
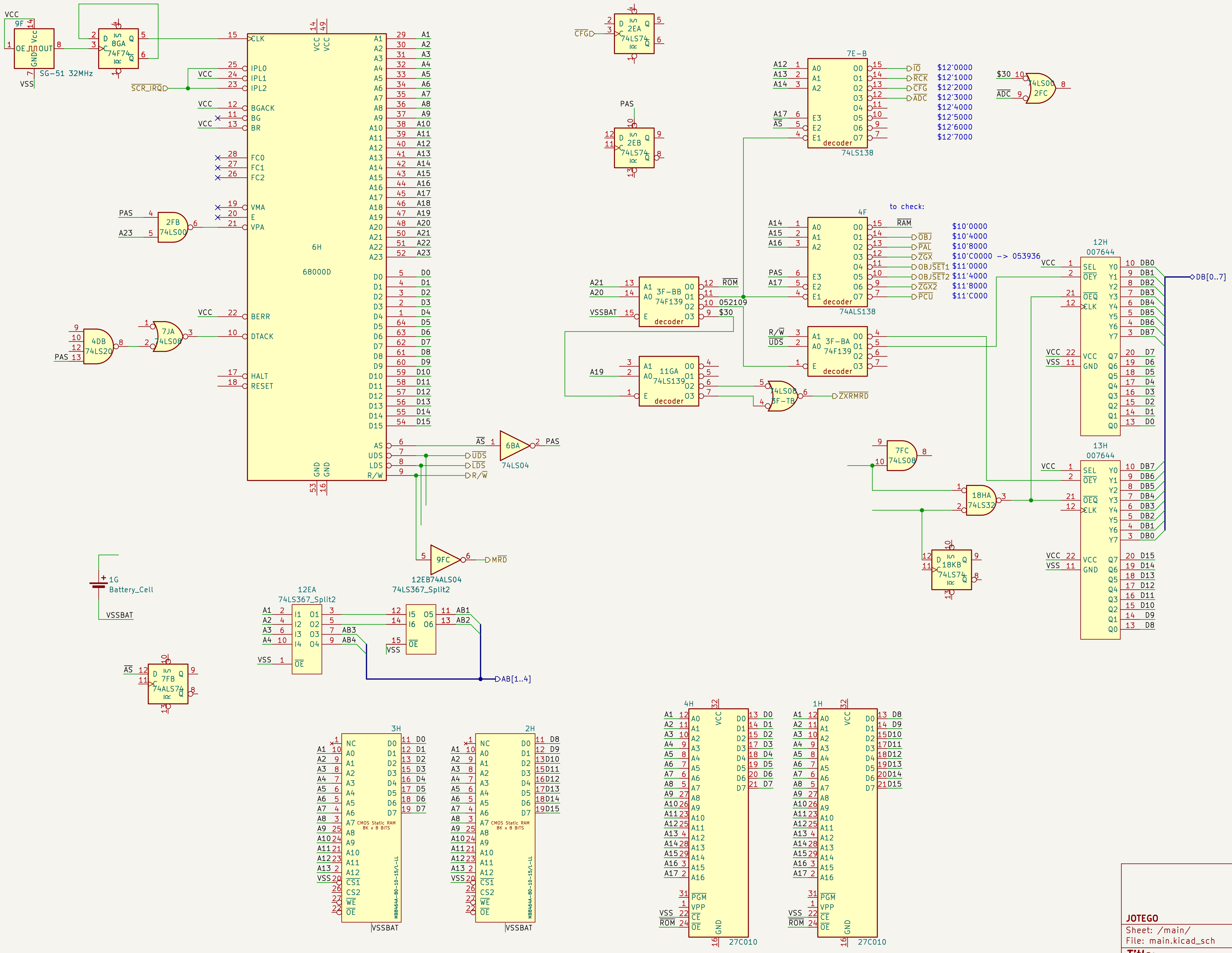

main
SCRIRQ
SCR_IRQ
UDS
LDS
MRD
R/W
AB[1..4]
DB[0..7]
CFG
ZXRMRD
IO
ADC
RCK
OBJREG
PAL
OBJ
PCU
ZGX
ZGX2
MRD
DB[0..7]
ZXRMRD
IO
ADC
RCK
File: main.kicad_sch
sound
File: sound.kicad_sch
scroll
SCRCLK
MAIN_A[15..1]
PQ
SYSRES
UDS
M12
VRAM
RESET
MRD
M6
RMRD
RMRD
IRQ
SCRIRQ
APAL[2..0]
ACOL[3..0]
MAIN_DB[7..0]
FPAL[2..0]
BPAL[2..0]
BCOL[3..0]
FCOL[3..0]
File: scroll.kicad_sch
zlayer
DB[0..7]
RCK
MRD
ZXRMRD
DB[0..7]
RCK
MRD
ZXRMRD
File: zlayer.kicad_sch
objects
File: objects.kicad_sch
cabinet
CFG
ADC
MRD
MWR
IO
CFG
ADC
MRD
MWR
IO
ZFX_BANK
RMRD
RMRD
DB[0..15]
ZFXTST[0..1]
File: cabinet.kicad_sch

JOTEGO
Sheet: /
File: glfgreat.kicad_sch

Title:

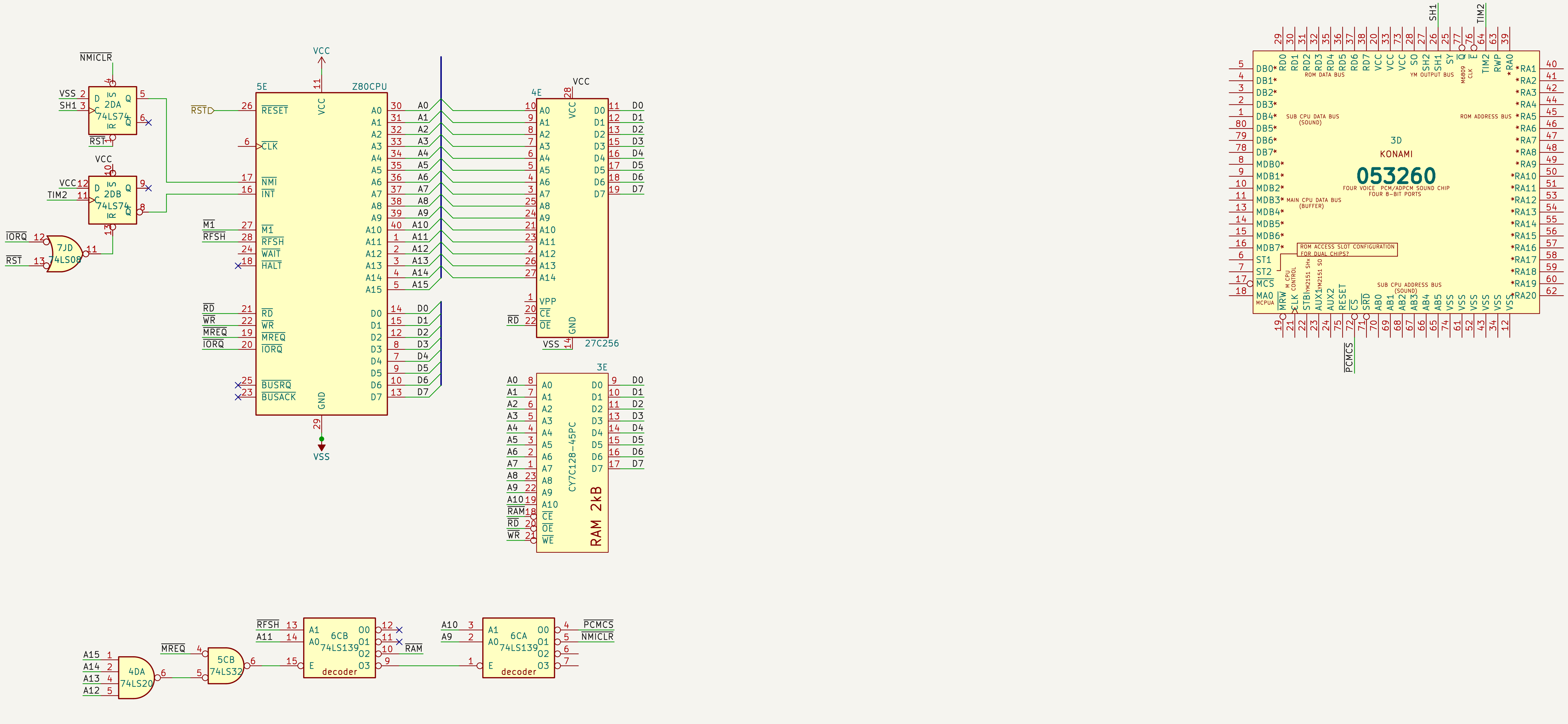
Size: A3
Date: 2025-06-21
Rev:
KiCad E.D.A. 8.0.5
Id: 1/7


SG-51 32MHz
9F
8GA
74F74
6H
68000D
CLK
IPL0
IPL1
IPL2
SCR_IRQ
BGACK
BG
BR
FC0
FC1
FC2
VMA
E
VPA
BERR
DTACK
HALT
RESET
2FB
74LS00
7JA
74LS08
4DB
74LS20
PAS
A23
VCC
VSS
AS
UDS
LDS
R/W
6BA
74LS04
9FC
12EB74ALS04
MRD
12EA
74LS367_Split2
74LS367_Split2
AB1
AB2
AB3
AB4
AB[1..4]
1G
Battery_Cell
VSSBAT
7FB
74ALS74
2EA
74LS74
CFG
2EB
74LS74
7E-B
74LS138
decoder
IO
RCK
CFG
ADC
$12'0000
$12'1000
$12'2000
$12'3000
$12'4000
$12'5000
$12'6000
$12'7000
$30
ADC
2FC
74LS00
4F
74ALS138
to check:
RAM
OBJ
PAL
ZGX
OBJSET1
OBJSET2
ZGX2
PCU
$10'0000
$10'4000
$10'8000
$10'C0000 -> 053936
$11'0000
$11'4000
$11'8000
$11'C000
3F-BB
74F139
ROM
052109
$30
VSSBAT
A21
A20
A19
11GA
74LS139
3F-BA
74F139
3F-TB
74LS08
ZXRMRD
12H
007644
13H
007644
DB[0..7]
SEL
OEY
OEQ
7FC
74LS08
18HA
74LS32
18KB
74LS74
3H
2H
CMOS Static RAM 8K x 8 BITS
MB8464A-80-10-15/L-LL
4H
1H
27C010
PGM
VPP
CE
OE

JOTEGO
Sheet: /main/
File: main.kicad_sch

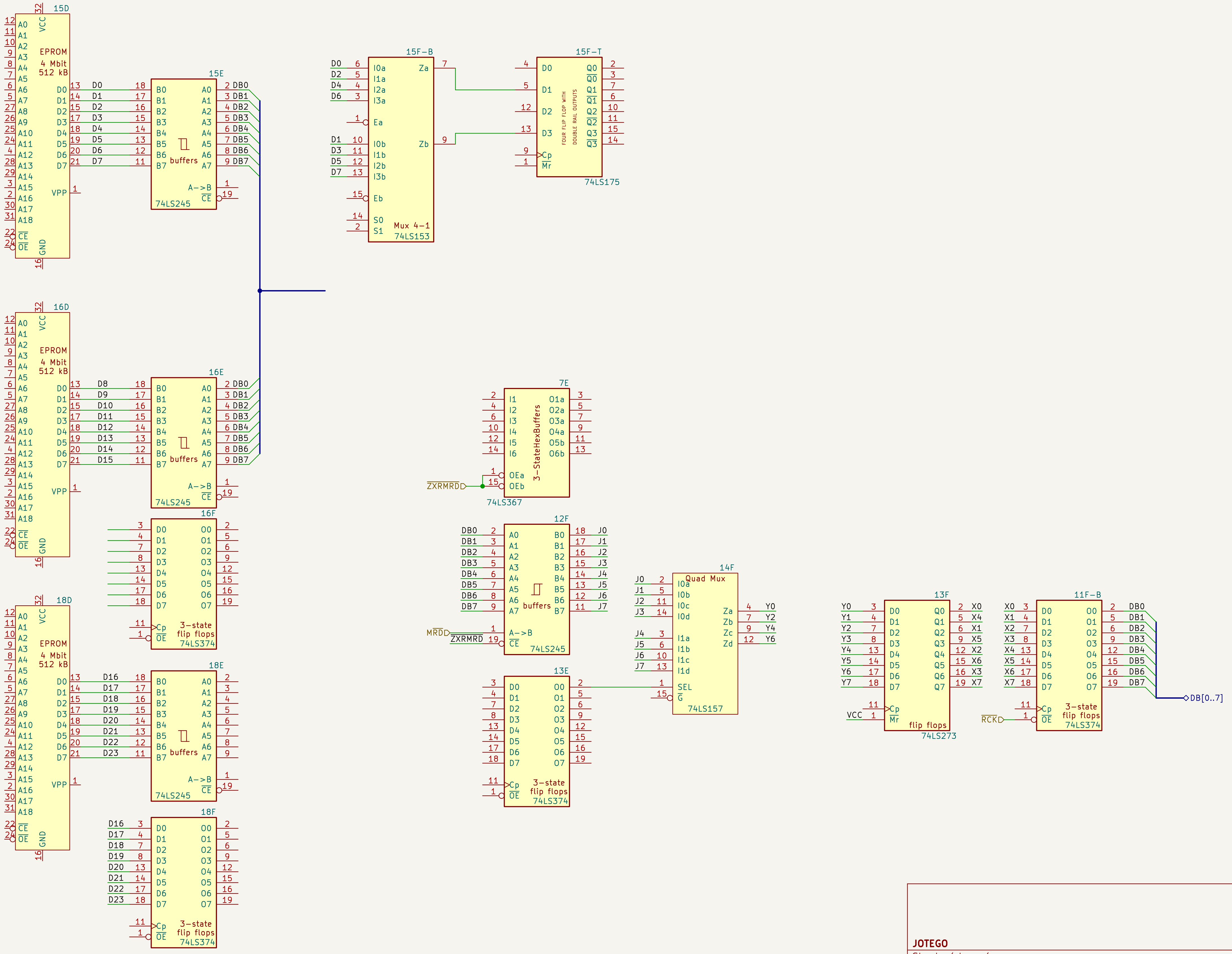
Title:

Size: A3 Date: 2025-06-21 **Rev:**
KiCad E.D.A. 8.0.5 Id: 2/7



NMICLR
VSS
SH1
2DA
74LS74
RST
VCC
TIM2
2DB
IORQ
RST
7JD
74LS08
RSTD
VCC
5E
Z80CPU
RESET
CLK
NMI
INT
M1
RFSH
WAIT
HALT
RD
WR
MREQ
IORQ
BUSRQ
BUSACK
GND
VSS
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
A14
A15
D0
D1
D2
D3
D4
D5
D6
D7
4E
27C256
VPP
CE
OE
3E
CY7C128-45PC
RAM 2kB
WE
RAM
A15
A14
A13
A12
4DA
74LS20
MREQ
5CB
74LS32
RFSH
A11
6CB
74LS139
decoder
A10
A9
6CA
74LS139
decoder
PCMCS
NMICLR
SH1
TIM2
3D
KONAMI
053260
FOUR VOICE PCM/ADPCM SOUND CHIP
FOUR 8-BIT PORTS
ROM DATA BUS
YM OUTPUT BUS
SUB CPU DATA BUS (SOUND)
ROM ADDRESS BUS
MAIN CPU DATA BUS (BUFFER)
ROM ACCESS SLOT CONFIGURATION FOR DUAL CHIPS?
SUB CPU ADDRESS BUS (SOUND)
DB0
DB1
DB2
DB3
DB4
DB5
DB6
DB7
MDB0
MDB1
MDB2
MDB3
MDB4
MDB5
MDB6
MDB7
ST1
ST2
MCS
MA0
MCPUA
RD0
RD1
RD2
RD3
RD4
RD5
RD6
RD7
SO
SH2
SY
TIM2
RWP
RA0
RA1
RA2
RA3
RA4
RA5
RA6
RA7
RA8
RA9
RA10
RA11
RA12
RA13
RA14
RA15
RA16
RA17
RA18
RA19
RA20
MRW
CLK
STBI
AUX1
AUX2
RESET
CS
SRD
AB0
AB1
AB2
AB3
AB4
AB5
VSS
PCMCS

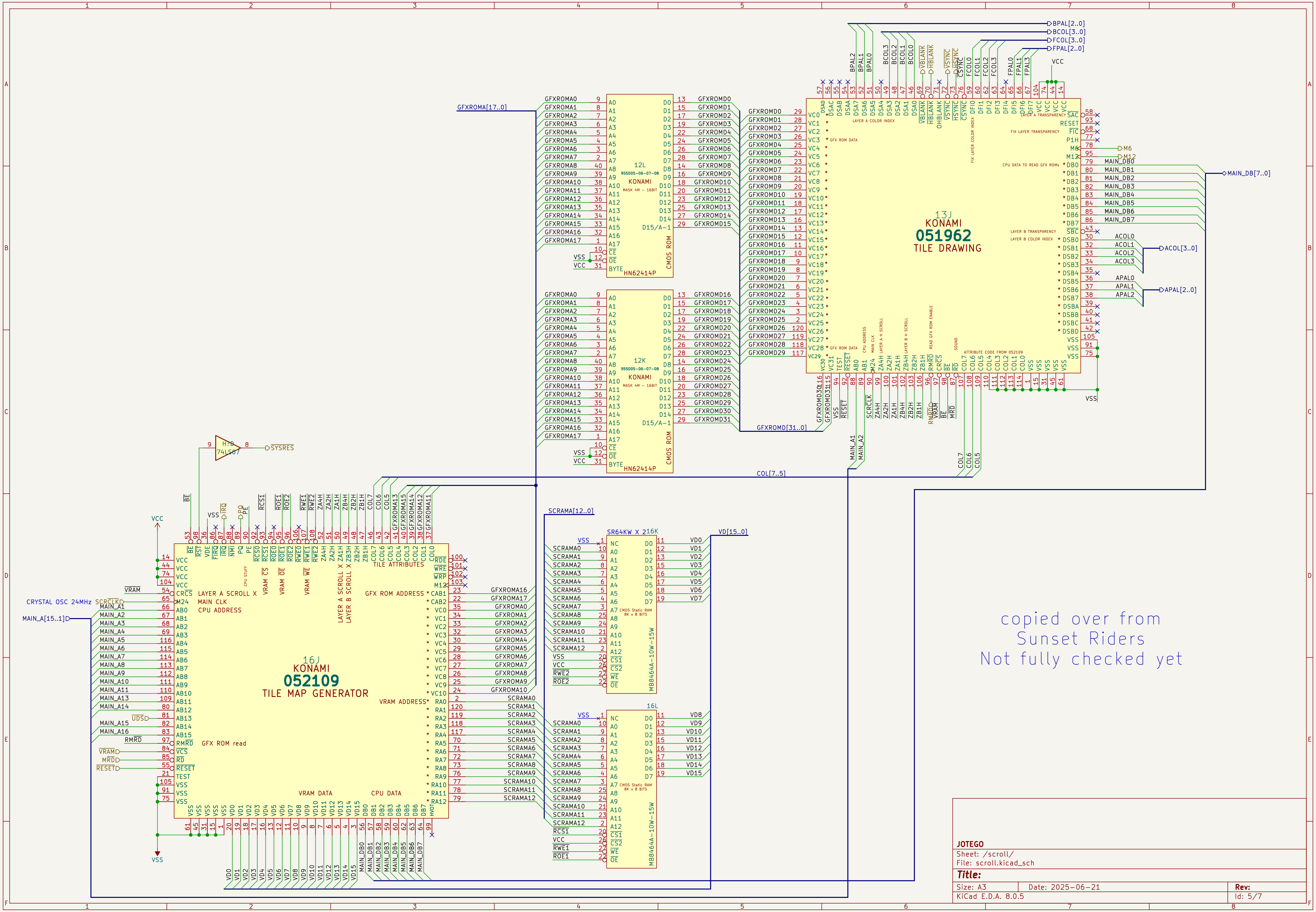
JOTEGO
Sheet: /sound/
File: sound.kicad_sch
Title:
Size: A3
Date: 2025-06-21
Rev:
KiCad E.D.A. 8.0.5
Id: 3/7


15D
EPROM 4 Mbit 512 kB
15E
buffers
74LS245
16D
EPROM 4 Mbit 512 kB
16E
buffers
74LS245
16F
3-state flip flops
74LS374
18D
EPROM 4 Mbit 512 kB
18E
buffers
74LS245
18F
3-state flip flops
74LS374
15F-B
Mux 4-1
74LS153
15F-T
FOUR FLIP FLOP WITH DOUBLE RAIL OUTPUTS
74LS175
7E
3-StateHexBuffers
74LS367
ZXRMRD
12F
buffers
74LS245
MRD
ZXRMRD
14F
Quad Mux
74LS157
13E
3-state flip flops
74LS374
13F
flip flops
74LS273
VCC
11F-B
3-state flip flops
74LS374
RCK
DB[0..7]

JOTEGO
Sheet: /zlayer/
File: zlayer.kicad_sch

Title:

Size: A3 | Date: 2025-06-21 | **Rev:**
KiCad E.D.A. 8.0.5 | Id: 4/7

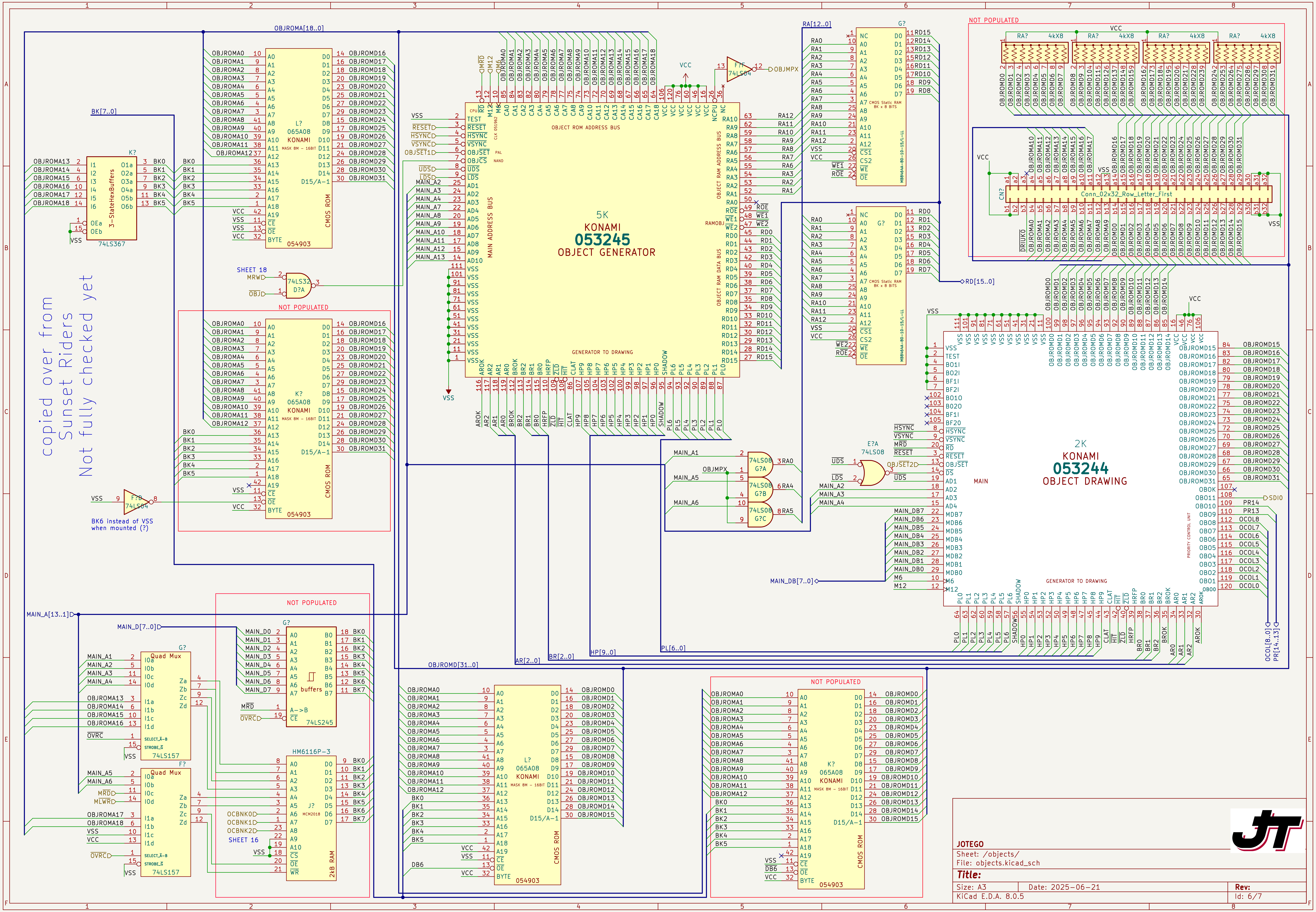

BPAL[2..0]
BCOL[3..0]
FCOL[3..0]
FPAL[2..0]
VCC
VSS
GFXROMA[17..0]
12L
955005-06-07-08
KONAMI
MASK 4M - 16BIT
HN62414P
CMOS ROM
12K
955005-06-07-08
KONAMI
MASK 4M - 16BIT
HN62414P
CMOS ROM
13J
KONAMI
051962
TILE DRAWING
LAYER A COLOR INDEX
GFX ROM DATA
LAYER A TRANSPARENCY
FIX LAYER TRANSPARENCY
FIX LAYER COLOR INDEX
CPU DATA TO READ GFX ROMs
LAYER B TRANSPARENCY
LAYER B COLOR INDEX
ATTRIBUTE CODE FROM 052109
READ GFX ROM ENABLE
SOUND
CPU ADDRESS
MAIN CLK
LAYER A H SCROLL
LAYER B H SCROLL
M6
M12
MAIN_DB[7..0]
ACOL[3..0]
APAL[2..0]
GFXROMD[31..0]
MAIN_A1
MAIN_A2
COL[7..5]
H:D
74LS07
SYSRES
16J
KONAMI
052109
TILE MAP GENERATOR
LAYER A SCROLL X
MAIN CLK
CPU ADDRESS
CPU STUFF
TILE ATTRIBUTES
GFX ROM ADDRESS
VRAM ADDRESS
GFX ROM read
VRAM DATA
CPU DATA
LAYER A SCROLL X ZA1H
LAYER B SCROLL X ZB3H
CRYSTAL OSC 24MHz
SCRCLK
MAIN_A[15..1]
VRAM
UDSD
RMRD
MRD
RESET
SCRAMA[12..0]
VD[15..0]
SR64KW X 2
16K
16L
CMOS Static RAM 8K x 8 BITS
MB8464A-10W-15W

copied over from
Sunset Riders
Not fully checked yet

JOTEGO
Sheet: /scroll/
File: scroll.kicad_sch

Title:

Size: A3 | Date: 2025-06-21 | Rev:
KiCad E.D.A. 8.0.5 | Id: 5/7


copied over from
Sunset Riders
Not fully checked yet
OBJROMA[18..0]
BK[7..0]
RA[12..0]
RD[15..0]
MAIN_A[13..1]
MAIN_D[7..0]
MAIN_DB[7..0]
OBJROMD[31..0]
AR[2..0]
BR[2..0]
HP[9..0]
PL[6..0]
OCOL[8..0]
PR[14..13]
OBJMPX
SDIO
5K
KONAMI
053245
OBJECT GENERATOR
OBJECT ROM ADDRESS BUS
MAIN ADDRESS BUS
OBJECT RAM ADDRESS BUS
OBJECT RAM DATA BUS
GENERATOR TO DRAWING
2K
KONAMI
053244
OBJECT DRAWING
PRIORITY CONTROL UNIT
MAIN
065A08
KONAMI
MASK 8M - 16BIT
CMOS ROM
054903
D15/A-1
3-StateHexBuffers
74LS367
74LS32
74LS04
74LS08
74LS245
buffers
74LS157
Quad Mux
HM6116P-3
MCM2018
2kB RAM
MB8464A-80-10-15/L-LL
CMOS Static RAM 8K x 8 BITS
NOT POPULATED
BK6 instead of VSS
when mounted (?)
SHEET 18
SHEET 16
Conn_02x32_Row_Letter_First
RA?
4kX8
VCC
VSS

JOTEGO

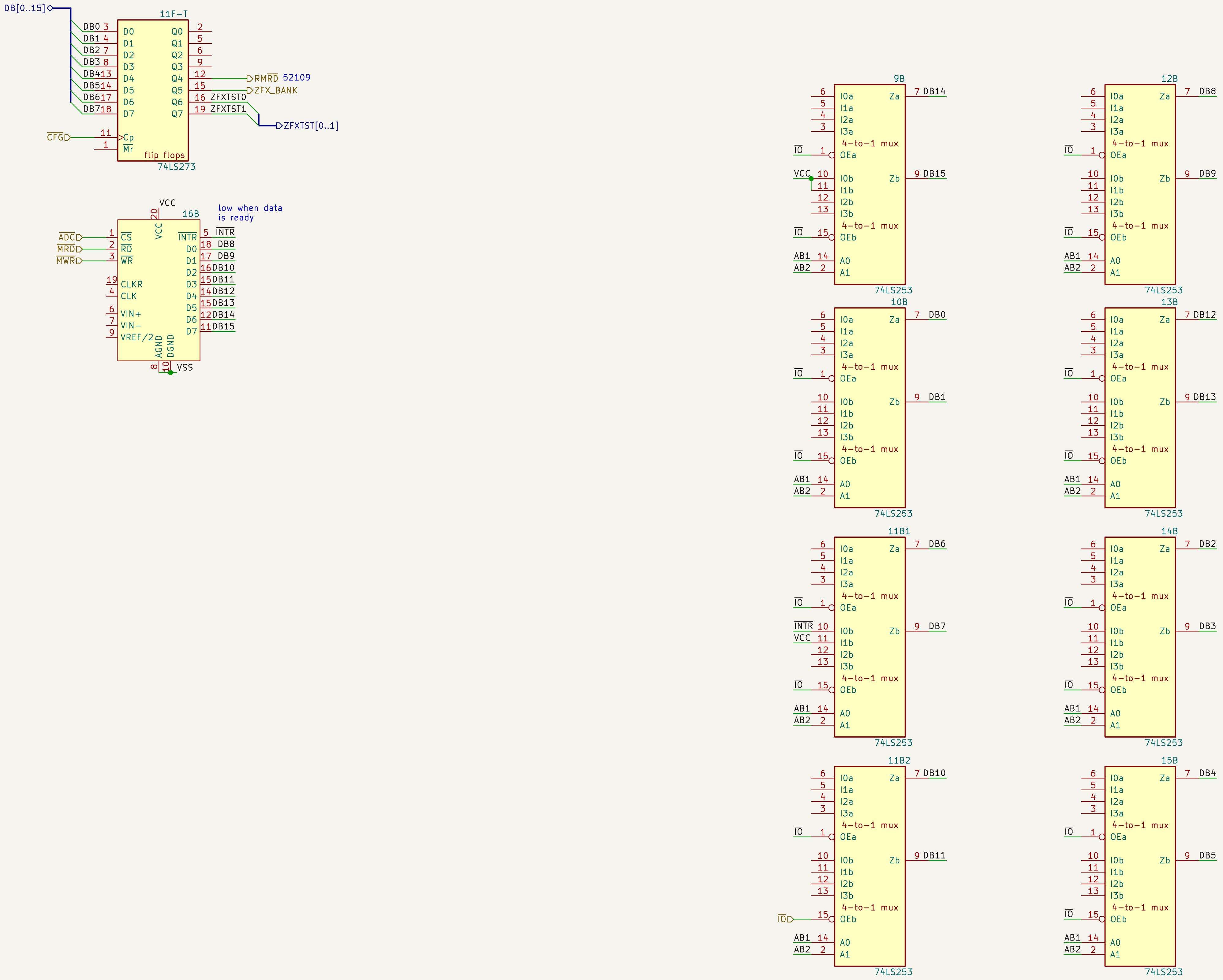
Sheet: /objects/

File: objects.kicad_sch

Title:

Size: A3 Date: 2025-06-21 Rev:

KiCad E.D.A. 8.0.5 Id: 6/7


DB[0..15]
11F-T
DB0
DB1
DB2
DB3
DB4
DB5
DB6
DB7
D0
D1
D2
D3
D4
D5
D6
D7
Q0
Q1
Q2
Q3
Q4
Q5
Q6
Q7
CFG
Cp
Mr
flip flops
74LS273
RMRD 52109
ZFX_BANK
ZFXTST0
ZFXTST1
ZFXTST[0..1]
VCC
16B
low when data
is ready
ADC
MRD
MWR
CS
RD
WR
CLKR
CLK
VIN+
VIN-
VREF/2
AGND
DGND
VSS
INTR
DB8
DB9
DB10
DB11
DB12
DB13
DB14
DB15
9B
12B
10B
13B
11B1
14B
11B2
15B
I0a
I1a
I2a
I3a
4-to-1 mux
OEa
I0b
I1b
I2b
I3b
OEb
A0
A1
Za
Zb
IO
AB1
AB2
74LS253

JOTEGO

Sheet: /cabinet/
File: cabinet.kicad_sch

Title: Cabinet input/output

Size: A3
Date: 2025-06-21
Rev:
KiCad E.D.A. 8.0.5
Id: 7/7